

DIMENSIONALITY REDUCTION TECHNIQUES FOR FUNCTIONAL CONNECTOME FINGERPRINTING

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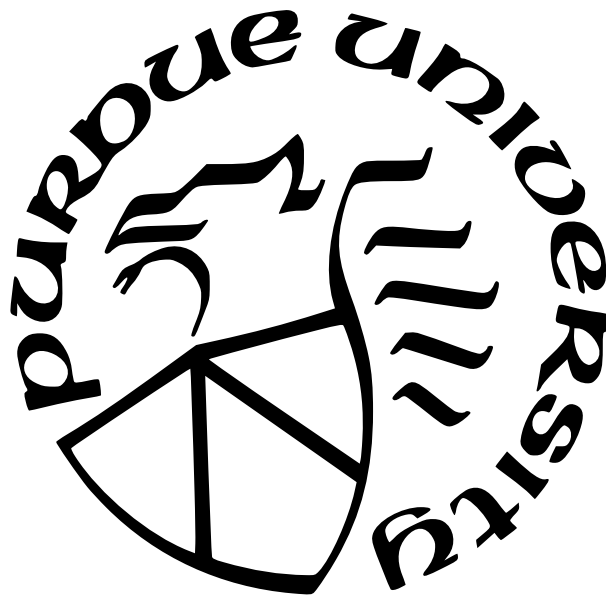
Vitor Farias Costa de Carvalho

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**THE PURDUE UNIVERSITY GRADUATE SCHOOL
STATEMENT OF COMMITTEE APPROVAL**

Dr. Ana María Estrada Gómez, Co-Chair

Edwardson School of Industrial Engineering

Dr. Joaquín Goñi Cortes, Co-Chair

Edwardson School of Industrial Engineering

Dr. Mario Ventresca

Edwardson School of Industrial Engineering

Approved by:

Dr. Young-Jun Son

To God be the glory, power, and honor.

To my family, advising committee, friends and colleagues.

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1. SUMMARY

Functional connectome (FC) fingerprinting is a rapidly evolving field within neuroimaging that seeks to identify unique patterns of functional connectivity in the human brain. The ability to reliably distinguish individuals based on their functional connectome has significant implications for neuroscience, particularly in understanding cognitive variability, mental disorders, and personalized medicine. Traditional approaches to FC fingerprinting rely on direct comparisons of functional connectivity matrices, but these methods can be limited by noise, high dimensionality, and the inherent variability of fMRI data across scanning sessions. To address these challenges, this thesis explores two distinct approaches utilizing dimensionality reduction techniques to enhance FC fingerprinting.

The first chapter investigates the use of tensor decomposition to improve the identifiability of individual functional connectivity patterns across multiple fMRI conditions. By decomposing high-dimensional FCs data into a core tensor and factor matrices, the framework successfully enhances the stability and distinctiveness of functional connectome fingerprints. The study examines within-condition and between-condition fingerprinting, demonstrating that Tucker decomposition significantly increases matching rates compared to methods that do not model the high-dimensionality of functional connectivity data, particularly for lower parcellation granularities.

The second chapter focuses on the application of Riemannian geometry-based methods, particularly tangent space projection, to further refine functional connectome fingerprints when several data acquisition sessions are available. This approach addresses the limitations of traditional Euclidean-based analyses by incorporating the geometric space in which FCs lie mapping FC matrices onto a tangent space, enabling robust comparisons while preserving individual-specific patterns. By leveraging Pearson correlation and Euclidean distance metrics, the study highlights the advantages of tangent FCs in improving subject distinctiveness across multiple scanning sessions. The integration of t-SNE visualization further validates the effectiveness of this method in clustering subject-specific connectivity patterns.

2. FUNCTIONAL CONNECTOME FINGERPRINTING THROUGH TUCKER TENSOR DECOMPOSITION

2.1 Introduction

Human brain connectomics, driven by the increasing availability of large-scale neuroimaging datasets [1], has emerged in recent years as a prominent field of research. This field has the potential to address many of the open questions about the structure and function of the human brain. Notably, connectomics-based analyses have revealed meaningful differences between healthy and disease conditions [2, 3]. However, to further assess the reliability of such findings and to capture individual-specific characteristics that may be overlooked in group-level studies, the concept of a “brain connectivity fingerprint” has gained growing interest [4–6].

Functional connectivity fingerprinting of the brain refers to the ability to identify an individual’s functional connectome (FC) from a set of FCs in repeated fMRI imaging sessions. The existence of a brain fingerprint has been established in the last decade with work done with data from functional magnetic resonance imaging (fMRI) and electroencephalography (EEG) [7, 8]. Such studies have shown that the functional connectome of the brain varies between individuals, therefore serving to some extent as a *fingerprint*. In the literature, different studies of FC fingerprinting have been conducted with varying approaches, such as principal component analysis (PCA) [9, 10], sparse dictionary learning (SDL) [11], geodesic distance in regularized FC [12] and correlation distance in FC tangent space projections [13]. In [9], the authors show that individual connectivity profiles can be reconstructed through an optimal linear combination of PCA-derived orthogonal components. In [10], the authors perform PCA in a subset of “learning” FCs to obtain an eigenspace to which the “validation” FCs are projected, thus enabling the identification of the fMRI condition or the participant to which the “validation” FC belongs. In [11], SDL is used to refine FC profiles, leading to a higher distinctiveness in FCs relative to raw connectivity. PCA and SDL work on 2-dimensional data, while Tucker decomposition is designed for higher-dimensional data structures called tensors, essentially allowing it to analyze complex relationships across multiple variables in a dataset by decomposing it into a core tensor and factor matrices

along each dimension. In essence, Tucker decomposition may be thought of as a "higher-order PCA".

Using the fact that FCs estimated as correlation matrices lie on or inside a Symmetric Positive Definite (SPD) manifold, Venkatesh and colleagues proposed using geodesic distance to compare FCs [14]. In a follow-up study [12], the authors explored how optimally regularized FCs maximize the individual fingerprint of participants, measured by the geodesic distance between their FCs. A limitation of this approach is that the geodesic distance between FCs of size $M \times M$ is a map $\mathbb{R}^{M \times M} \times \mathbb{R}^{M \times M} \rightarrow \mathbb{R}$. Hence, even though the geodesic distance provides a global measure of similarity between FCs, it does not directly highlight the specific features that make the individual's functional connectivity unique. As an alternative to using FCs, tangent FCs have demonstrated a high capacity to predict cognition and behavior [15–17]. Most recently, [13] analyzed the effects of tangent FCs with respect to fingerprinting. In that study, a high degree of fingerprinting was achieved not only across sessions of a unique participant, but also for matching the sessions of twins.

To simultaneously overcome the drawbacks of the studies mentioned above, we propose to utilize tensor decomposition for FC fingerprinting. Tensor decomposition enables projecting high-dimensional data into a lower-dimensional space, preserving its structure while independently extracting meaningful information from each dimension.

Tensors are multidimensional arrays with applications in various fields, including signal processing, computer vision, and neuroscience [18]. In brain connectomics, tensors enable modeling and analyzing the functional and structural connections within the brain by reducing the dimensionality of complex, interrelated, and high-dimensional data through tensor decomposition. In [19], the authors studied the dynamics of FCs to understand the process of formation and dissolution of brain functional networks through tensor decomposition techniques. In another study [20], it was demonstrated how the analysis of the tensor components enables the extraction of unbiased and interpretable descriptions of single-trial dynamics across many trials through low-dimensional representations of neural data. In [21], the authors discuss challenges associated with interpreting the brain connectivity patterns derived from tensor decomposition. To our knowledge, only [22] has considered using tensors in brain connectivity fingerprinting analysis in the literature. However, their study was

based on structural connectivity. The task of identifying subjects through their functional connectomes has the additional challenge of dynamic changes and functional reconfigurations happening at a fast rate in response to cognitive stimuli.

Through this chapter, the effectiveness of tensor based methods in uncovering FC fingerprints is assessed. The adopted fingerprinting framework can be broadly described as follows:

1. Each participant underwent a total of sixteen data acquisition sessions, two for each of the considered fMRI conditions. Using the BOLD time series from each acquisition, an FC matrix of dimension Number of Brain Regions $\times$ Number of Brain Regions is estimated, thus yielding a total of sixteen FC matrices for each participant.
2. For each condition, a tensor of dimensions Number of Brain Regions $\times$ Number of Brain Regions $\times$ Number of Participants, is constructed by concatenating all participants' FCs derived in the first acquisition session. Similarly, a second tensor is obtained for all conditions by concatenating all participants' FCs derived in the second acquisition session.
3. The obtained tensors are then decomposed via Tucker decomposition [23], yielding a core tensor and three factor matrices. The first two factor matrices contain cohort-level functional connectivity information and will later on be referred to as "brain parcellation factor matrices". The third factor matrix contains participant-specific information and will be later addressed as the "participants factor matrix". The participants factor matrix, acting as a "fingerprint" of each participant, is used to match participants FCs corresponding to different data acquisition sessions. The accuracy with which different sessions are able to be correctly matched is quantified by a metric denominated matching rate [24].

The aims of this chapter are: (i) assess the impact of Tucker decomposition on functional connectome fingerprinting in within- and between-fMRI condition settings for different parcellation granularities, (ii) estimate optimal levels of compression of brain parcellation-specific and participant-specific information that maximize fingerprinting, and (iii) analyze how sampling in resting-state prior to Tucker decomposition affects fingerprinting.

The remainder of the chapter is organized as follows. In the [Materials and Methods](#) section, we: (i) describe the data set used in this chapter as well as the preprocessing procedures, (ii) introduce the adopted tensor notation, (iii) introduce Tucker decomposition, and (iv) describe matching rate, the fingerprinting measurement used in this chapter. In the [Results](#) section, we: (i) provide a comparative analysis of fingerprints across parcellation granularities, (ii) compare the fingerprinting performance of different dimensionality reduction methods across parcellation granularities, (iii) disclose optimal levels of compression of parcellation-specific and participant-specific information that maximize within- and between-condition fingerprints, and (iv) present the findings for the different strategies of sampling time points in resting-state time series. In the [Discussion](#) section, we (i) discuss our findings and (ii) highlight some limitations of our study and make suggestions for future work. Lastly, in the [Conclusion](#) section we summarize the presented work.

2.2 Materials and Methods

In this section, a description of the dataset, tensor notation, and fingerprinting framework is presented. First, an overview of the HCP dataset, brain parcellation granularities considered, and preprocessing procedures is presented. Then, the adopted tensor notation is introduced, along with commonly used tensor decomposition techniques. Lastly, we introduce the metric used to quantify FC fingerprinting, and a detailed description of the fingerprinting framework.

2.2.1 Dataset Description

The HCP Dataset [25] has been widely used as a standard in the neuroimaging literature for a broad range of research domains [7, 26–29] due to its large-scale, high-quality, and open-access data gathered from a large and diverse cohort of participants. This chapter utilizes data from eight fMRI conditions included in the Young-Adult Human Connectome Project (HCP) dataset. To minimize the impact of hereditary influences on fingerprinting, a subset of 426 unrelated individuals was selected. The demographic information of the participants can be found in Table 2.1. The fMRI conditions analyzed in this chapter include resting-state

(RS), Emotion processing , Gambling (GAM), Language (LAN), Motor (MOT), Relational processing (REL), Social cognition (SOC), and Working Memory (WM). Each participant completed two sessions per condition, corresponding to separate left-to-right (LR) and right-to-left (RL) acquisitions, which are designated as test and retest sessions. Resting-state scans were conducted over four sessions ("REST1" and "REST2") on two separate days, though only the two REST1 sessions were considered in this chapter.

Table 2.1. Demographic information of the 426 unrelated participants from HCP’s Young Adult Dataset.

Demographic Information ($n = 426$, 203 males)	Mean (SD)	Range
Age	28.67 (3.78)	22-36
Years of Education	14.99 (1.77)	11-17

2.2.2 Brain Parcellations

In this chapter, we utilized the Schaefer parcellation functional brain atlas to analyze the human cortex [30]. This parcellation was derived from resting-state fMRI data of 1,489 participants, and registered with surface alignment. It was generated using a gradient-weighted Markov random field approach that integrates local gradient and global similarity methods. The Schaefer parcellation is available at nine levels of granularity, ranging from 100 to 900 regions in increments of 100, in both volumetric and grayordinate space. Since the grayordinate versions of these parcellations are already in the same surface space as the HCP fMRI data, mapping them onto the fMRI scans is straightforward. Moreover, surface-based mapping ensures better alignment between the parcellations and fMRI data compared to volumetric mapping. Therefore, we applied surface-based mapping to project the Schaefer parcellations onto the fMRI data. To ensure comprehensive brain coverage, we incorporated 14 subcortical regions into each parcellation, as provided in the HCP dataset (file: Atlas_ROI2.nii.gz). This file was converted from NIFTI to CIFTI format using the HCP Workbench software (`wb_command -cifti-createlabel`). As a result, the Schaefer-200 parcellation, for instance, ultimately included 214 brain regions.

2.2.3 Preprocessing

The preprocessing of fMRI data followed the "minimal" preprocessing pipeline provided by the HCP, which includes artifact removal, motion correction, and alignment to a standardized template [31]. Further details on this pipeline can be found in previous studies [26, 32]. We enhanced this minimal pipeline by incorporating additional preprocessing steps, as described in [13]. Specifically, for resting-state fMRI data, we applied the following procedures: (i) regressed out the global gray matter signal from voxel-wise time series [31], (ii) implemented a first-order Butterworth bandpass filter in both forward and reverse directions within the frequency range of 0.001-0.08 Hz [31], and (iii) z-scored and averaged voxel time series within each brain region, while excluding time points that deviated more than three standard deviations from the mean (processed using the Workbench software, `wb_command -cifti-parcellate`). The same preprocessing steps were applied to all task-based fMRI conditions. However, the bandpass filter was adjusted to a broader frequency range (0.001-0.250 Hz) [33], as the optimal filtering range remains uncertain [34].

2.2.4 Estimation of whole-brain FCs

The functional connectivity between pairs of brain regions was estimated by computing Pearson's correlation (*corr* MATLAB function), which results in a symmetric $M \times M$ correlation matrix, with M being the number of brain regions for a given parcellation. Throughout this chapter, this correlation matrix is referred to as FC. For each participant, we computed a whole-brain FC for each of the two sessions (test and retest), each fMRI condition (all seven tasks and resting state), and all parcellation granularities.

2.2.5 Tensor Notation

We refer to multidimensional data structures as tensors. Mathematically, tensors can be described as objects that lie in $\mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ and originate from the tensor product of N vector spaces. The number N of vector spaces from which a tensor originates defines the order (or mode) of the tensor. Throughout this chapter, we adopt the following notation:

scalars are denoted by lower or upper case letters, e.g., x or X , vectors are denoted by boldface lowercase letters, e.g., $\mathbf{x}$, matrices are denoted by boldface capital letters, e.g., $\mathbf{X}$, and high-order tensors are denoted by boldface calligraphic letters, e.g., $\mathcal{X}$. Entries of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ are denoted by lowercase letters with subscripts, e.g. $x_{i_1, i_2, \dots, i_N}$, with $1 \leq i_n \leq I_n$ for all $1 \leq n \leq N$. Tensor fibers are constructed by fixing every index of the tensor but one. Therefore, tensors have as many fibers as dimensions. For example, third-order tensors have column (mode-1), row (mode-2), and tube (mode-3) fibers, which are denoted by $\mathbf{x}_{:jk}$, $\mathbf{x}_{i:k}$, and $\mathbf{x}_{ij:}$, respectively, with colons denoting all the entries of a mode. A tensor slice is defined by fixing all entries from the tensor except two. We can define the slices $\mathbf{X}_{i:, :}$, $\mathbf{X}_{:, j, :}$, and $\mathbf{X}_{:, :, k}$ for a third-order tensor.

The process of reshaping a tensor into a matrix is known as matricization. The mode- n matricization of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ is given by the matrix $\mathbf{X}_{(n)}$, and its columns correspond to the mode- n fibers of $\mathcal{X}$. The n -mode matrix product between a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ and a matrix $\mathbf{Y} \in \mathbb{R}^{J \times I_n}$ is denoted by $\mathcal{X} \times_n \mathbf{Y}$ and is of size $I_1 \times \dots \times I_{n-1} \times J \times I_{n+1} \times \dots \times I_N$. The outer product between two vectors $\mathbf{x} \in \mathbb{R}^m$ and $\mathbf{y} \in \mathbb{R}^n$ is represented by

$$\mathbf{x} \circ \mathbf{y} = \begin{bmatrix} x_1 y_1 & x_1 y_2 & \cdots & x_1 y_n \\ x_2 y_1 & x_2 y_2 & \cdots & x_2 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_m y_1 & x_m y_2 & \cdots & x_m y_n \end{bmatrix}$$

and is of dimension $\mathbb{R}^{m \times n}$, while the inner product of $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ and $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ is defined as

$$\langle \mathcal{X}, \mathcal{Y} \rangle = \sum_{i_1=1}^{I_1} \sum_{i_2=1}^{I_2} \cdots \sum_{i_N=1}^{I_N} x_{i_1, i_2, \dots, i_N} y_{i_1, i_2, \dots, i_N}$$

and the norm of $\mathcal{X}$ is defined as

$$\|\mathcal{X}\| = \sqrt{\langle \mathcal{X}, \mathcal{X} \rangle}.$$

2.2.6 Tensor Decomposition

In recent years, tensors have become increasingly popular in the fields of signal processing, machine learning, and neuroscience for their capacity to model complex high-order relation-

ships among objects [35–37]. Tensor decomposition enables projecting high-dimensional data into a lower-dimensional space while preserving the original structure of the data. For the purpose of brain fingerprinting, tensor decomposition has the potential of extracting unique features from each participant’s fMRI data acquisition session, thus facilitating subject distinctiveness. Several tensor decomposition algorithms can be found in the literature, each with their own characteristics and applications. The most commonly used ones are the CANDECOMP/PARAFAC (CP) [38, 39] decomposition and the Tucker decomposition [23].

The CP decomposition of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ factorizes it into a sum of R rank-one tensors, where a rank-one tensor denotes the outer product between N vectors. Equation 2.1 shows the CP decomposition of $\mathcal{X}$:

$$\mathcal{X} \approx \sum_{r=1}^R \lambda_r \mathbf{a}_r^{(1)} \circ \mathbf{a}_r^{(2)} \dots \circ \mathbf{a}_r^{(N)} \quad (2.1)$$

where R denotes the rank of the decomposition. The vectors $\mathbf{a}_r^{(1)} \in \mathbb{R}^{I_1}, \mathbf{a}_r^{(2)} \in \mathbb{R}^{I_2}, \dots, \mathbf{a}_r^{(N)} \in \mathbb{R}^{I_N}$ are typically assumed to be normalized, with a corresponding scaling factor of λ_r . Equation 2.1 can also be expressed in the simplified form $\llbracket \boldsymbol{\lambda}; \mathbf{A}^{(1)}, \mathbf{A}^{(2)}, \dots, \mathbf{A}^{(N)} \rrbracket$, in which $\mathbf{A}^{(i)} = [\mathbf{a}_1^{(i)} \quad \mathbf{a}_2^{(i)} \quad \dots \quad \mathbf{a}_R^{(i)}]$, for $i = 1, 2, \dots, N$, are referred to as factor matrices, and $\boldsymbol{\lambda} \in \mathbb{R}^R$ is a vector containing all scaling factors λ_r , for $r = 1, 2, \dots, R$.

The Tucker decomposition [23] decomposes a tensor into a core tensor multiplied by a matrix along each of the tensor modes. For $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$, its n -rank, denoted by $rank_n(\mathcal{X})$, is defined as the column rank of its mode- n matricization $\mathbf{X}_{(n)}$. In other words, the n -rank is the number of linearly independent vectors that span the basis of the *mode- n* fibers of $\mathcal{X}$. Equation 2.2 shows the Tucker decomposition of $\mathcal{X}$.

$$\mathcal{X} \approx \sum_{r_1=1}^{R_1} \sum_{r_2=1}^{R_2} \dots \sum_{r_N=1}^{R_N} g_{r_1 r_2 \dots r_N} \mathbf{a}_{r_1}^{(1)} \circ \mathbf{a}_{r_2}^{(2)} \circ \dots \circ \mathbf{a}_{r_N}^{(N)} \equiv \llbracket \mathcal{G}; \mathbf{A}^{(1)}, \mathbf{A}^{(2)}, \dots, \mathbf{A}^{(N)} \rrbracket \quad (2.2)$$

where $\mathbf{A}^{(1)} \in \mathbb{R}^{I_1 \times R_1}, \mathbf{A}^{(2)} \in \mathbb{R}^{I_2 \times R_2}, \dots, \mathbf{A}^{(N)} \in \mathbb{R}^{I_N \times R_N}$ are column-wise orthonormal factor matrices, and $R_1, R_2, \dots, R_N$ are the ranks of the decomposition, where $R_n \leq rank_n(\mathcal{X}) \leq I_n$ for $n = 1, \dots, N$. If $R_n < rank_n(\mathcal{X})$, we refer to the decomposition as a truncated

Tucker decomposition and refer to it as a rank- $(R_1, R_2, \dots, R_N)$ decomposition. The tensor $\mathcal{G} \in \mathbb{R}^{R_1 \times R_2 \times \dots \times R_N}$ is referred to as the core tensor and its entries represent the level of interaction between the different factors. Note that the CP decomposition can be understood as a special case of the Tucker decomposition when the Tucker's core is reduced to a hyper-diagonal tensor (all non-diagonal entries are equal to zero) and $R = R_1 = R_2 = \dots = R_N$. For simplicity, consider the third-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$. For simplicity, consider the third-order tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$. The Tucker decomposition of $\mathcal{X}$ is obtained by solving the optimization problem 2.3, which seeks to minimize the norm of the difference between the true and estimated tensors. The decision variables of this problem are the core tensor $\mathcal{G}$, and the factor matrices $\mathbf{A}^{(1)}$, $\mathbf{A}^{(2)}$, and $\mathbf{A}^{(3)}$.

$$\begin{aligned}
& \min_{\mathcal{G}, \mathbf{A}^{(1)}, \mathbf{A}^{(2)}, \mathbf{A}^{(3)}} \quad \left\| \mathcal{X} - \mathcal{G} \times_1 \mathbf{A}^{(1)} \times_2 \mathbf{A}^{(2)} \times_3 \mathbf{A}^{(3)} \right\| & (2.3) \\
& \text{s.t.} \quad \mathcal{G} \in \mathbb{R}^{R_1 \times R_2 \times R_3} \\
& \quad \mathbf{A}^{(1)} \in \mathbb{R}^{I_1 \times R_1}, \quad \text{column-wise orthonormal} \\
& \quad \mathbf{A}^{(2)} \in \mathbb{R}^{I_2 \times R_2}, \quad \text{column-wise orthonormal} \\
& \quad \mathbf{A}^{(3)} \in \mathbb{R}^{I_3 \times R_3}, \quad \text{column-wise orthonormal}
\end{aligned}$$

2.2.7 Tucker Decomposition of Functional Connectomes

For each of the eight fMRI conditions analyzed in this chapter, we represent the data as a third-order tensor $\mathcal{X} \in \mathbb{R}^{M \times M \times N}$. Here, M is equal to the granularity of the brain parcellation and $N = 426$ corresponds to the total number of participants. Given the symmetry of FC matrices, we refer to $\mathcal{X}$ as a semi-symmetric tensor, meaning it remains invariant under permutation of two (or more) indices. In our case, $x_{i,j,k} = x_{j,i,k}$ for $k = 1, \dots, 426$. All analyses performed in this chapter have an input of a semi-symmetric tensor $\mathcal{X}$ constructed by concatenating participants' FCs obtained in one fMRI scanning session (either test or retest).

Once the data have been structured as a semi-symmetric tensor, we can produce a low-rank estimation of the FCs through either of the previously mentioned tensor decomposition methods. Due to the lack of interactions between components, the results of CP decomposition are generally easier to interpret [21] compared to Tucker decomposition. However, this lack of interaction often leads CP to produce less accurate approximations of the original tensor, as measured by the L_2 norm. In contrast, Tucker decomposition leverages its core tensor to capture interactions between components, enabling it to approximate the original tensor with greater precision [40]. Considering the interpretability/accuracy trade-off in the context of brain fingerprinting, we focus on Tucker decomposition.

Several methods have been developed to estimate the Tucker decomposition. Notably, we highlight Sequentially Truncated Higher-Order Singular Value Decomposition (ST-HOSVD) [41, 42] and Higher-Order Orthogonal Iteration (HOOI) [43]. For a given tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$, ST-HOSVD sequentially computes the factor matrices via truncated eigenvalue decompositions of the mode- n matricizations (for $n = 1, 2, \dots, N$) of $\mathcal{X}$, while iteratively updating $\mathcal{X}$ at each step. The output of the algorithm is a core tensor and a set of column-wise orthogonal factor matrices. Similarly to PCA, the components from the factor matrices capture most of the variance across each of the tensor modes. The Tucker estimation via ST-HOSVD of $\mathcal{X}$ is shown in Algorithm 1.

Algorithm 1 Sequentially Truncated Higher-Order Singular Value Decomposition (ST-HOSVD)

Input: $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$, $\{R_1, R_2, \dots, R_N\}$

- 1: **for** $i \leftarrow 1$ to N **do**
- 2: Matricize the tensor along mode i to obtain $\mathbf{X}_{(i)}$
- 3: Compute the Gram matrix $\mathbf{Z}_{(i)} = \mathbf{X}_{(i)} \times \mathbf{X}_{(i)}^T$
- 4: Compute the eigenvalue decomposition of $\mathbf{Z}_{(i)}$ and sort the eigenvalues in descending order
- 5: $\mathbf{A}^{(i)} \leftarrow R_i$ eigenvectors corresponding to the sorted R_i eigenvalues of $\mathbf{Z}_{(i)}$
- 6: $\mathcal{X} \leftarrow \mathcal{X} \times_i \mathbf{A}^{(i)T}$
- 7: **end for**
- 8: $\mathcal{G} \leftarrow \mathcal{X}$

Output: $\mathcal{G} \in \mathbb{R}^{R_1 \times R_2 \times \dots \times R_N}$, $\mathbf{A}^{(1)} \in \mathbb{R}^{I_1 \times R_1}$, $\mathbf{A}^{(2)} \in \mathbb{R}^{I_2 \times R_2}$, $\dots$, $\mathbf{A}^{(N)} \in \mathbb{R}^{I_N \times R_N}$

On the other hand, HOOI utilizes an Alternating Least Squares (ALS) approach to estimate each of the factor matrices by sequentially solving sub-problems of the form in 2.4:

$$\begin{aligned} \max_{\mathbf{A}^{(n)}} \quad & \left\| \mathcal{X} \times_1 \mathbf{A}^{(1)} \times_2 \mathbf{A}^{(2)} \dots \times_N \mathbf{A}^{(N)} \right\| \\ \text{s.t.} \quad & \mathbf{A}^{(n)} \in \mathbb{R}^{I_n \times R_N}, \quad \text{column-wise orthonormal} \end{aligned} \tag{2.4}$$

where the factor matrices $\mathbf{A}^{(1)} \times_2 \mathbf{A}^{(2)} \dots \times_N \mathbf{A}^{(N)}$ are commonly initialized via the ST-HOSVD of $\mathcal{X}$. By iteratively optimizing each factor matrix while keeping the others fixed, HOOI provides a better fit as measured by the norm of the difference between the true and estimated tensors compared to ST-HOSVD, but for a higher computational cost. However, since HOOI is not guaranteed to converge to a global optimum nor to a stationary point [43, 44] and does not provide substantial fingerprinting improvements in comparison to ST-HOSVD (see 2.2), ST-HOSVD is the chosen algorithm to compute the Tucker decomposition of FCs.

When applied to tensors that exhibit partial (or full) symmetries, HOSVD preserves the symmetric structure of the tensor [41]. Hence, for a tensor consisting of one session (e.g., test sessions) of participants' FC matrices, the Tucker decomposition of $\mathcal{X} \in \mathbb{R}^{M \times M \times N}$ can be reformulated as:

$$\begin{aligned}
& \min_{\mathcal{G}, \mathbf{B}, \mathbf{P}} \quad \|\mathcal{X} - \mathcal{G} \times_1 \mathbf{B} \times_2 \mathbf{B} \times_3 \mathbf{P}\| & (2.5) \\
& \text{s.t.} \quad \mathcal{G} \in \mathbb{R}^{R_1 \times R_1 \times R_2} \\
& \quad \mathbf{B} \in \mathbb{R}^{M \times R_1}, \quad \text{column-wise orthonormal} \\
& \quad \mathbf{P} \in \mathbb{R}^{N \times R_2}, \quad \text{column-wise orthonormal}
\end{aligned}$$

where the factor matrices $\mathbf{B}$ and $\mathbf{P}$ obtained via HOSVD contain, respectively, brain parcellation and participant-specific information and the ranks R_1 and R_2 express the compression levels of brain parcellation and participant-specific information. While solving the optimization problem presented in 2.5, the brain parcellation ranks were chosen with a step size of $\frac{(\text{Parcellation} - 14)}{4}$. In addition to the previous parcellation ranks, we also performed a full-rank decomposition. Thus, for a parcellation granularity of 414, for example, the brain parcellation ranks were set to $R_1 = \{100, 200, 300, 400, 414\}$. In contrast, for all parcellation granularities explored, the participant ranks were set to $R_2 = \{50, 100, 150, 200, 250, 300, 350, 400, 426\}$.

Under the hypothesis that the functional connectivity patterns of a participant are, to some extent, reproducible across scanning sessions, we fix the core tensor $\mathcal{G}$ and brain parcellation factor matrix $\mathbf{B}$ derived from the Tucker decomposition of tensor $\mathcal{X}$ and estimate the participant factor matrix $\mathbf{Q}$ of the tensor $\mathcal{Y}$ comprising of FCs from another data acquisition session (e.g. retest session). By doing so, we aim to detect a consistent presence of underlying cohort-level functional connectivity patterns across different data acquisition sessions for each participant. The optimization problem shown in 2.6

$$\min_{\mathbf{Q}} \quad \|\mathcal{Y} - \mathcal{G} \times_1 \mathbf{B} \times_2 \mathbf{B} \times_3 \mathbf{Q}\| \quad (2.6)$$

and admits a closed-form solution given by

$$\mathbf{Q} = \mathbf{Y}_{(3)} \times [(\mathcal{G} \times_1 \mathbf{B} \times_2 \mathbf{B})_{(3)}]^\dagger$$

where † denotes the Moore-Penrose inverse [45] of a matrix, and $\mathbf{Y}_{(3)}$ and $(\mathcal{G} \times_1 \mathbf{B} \times_2 \mathbf{B})_{(3)}$ denote the *mode-3* matricization of $\mathcal{Y}$ and $(\mathcal{G} \times_1 \mathbf{B} \times_2 \mathbf{B})$, respectively.

2.2.8 Fingerprinting quantification

To quantify fingerprinting, we used a measure denominated matching rate [24] for an identifiability matrix $\mathbf{I} \in \mathbb{R}^{N \times N}$, where $i_{j,k}$ denotes the Pearson's correlation between the j -th row of the participant factor matrix $\mathbf{P}_{j,:}$, and the k -th row of the participant factor matrix $\mathbf{Q}_{k,:}$. The main diagonal entries of $\mathbf{I}$ represent similarity levels between different imaging sessions of the same participant. By hypothesis, we expect those entries to be higher than the off-diagonal entries, which represent the similarity level between different imaging sessions of different participants. Matching rate is a variation of ID_{rate} [7] that accounts for the fact that each participant is present only once in the test and the retest sets. ID_{rate} (2.7) is the average frequency at which a participant's test session is most highly correlated to their retest session, and their retest session is most highly correlated to their test session (note that one does not necessarily imply the other). For matching rates, we impose that once a test session is paired with a retest session, it can no longer be chosen for a new pairing. The relative frequency of successful participants matching in both directions is then averaged, yielding a value in the range $[0, 1]$, where 0 indicates a failure to correctly match any of the participant's FCs, and 1 indicates success in matching all participant's FCs correctly. An algorithmic description of the computation of the matching rate is presented in Algorithm 2.

$$ID_{rate} = \frac{\frac{\text{Number of correctly paired test-retest samples}}{\text{Total number of participants}} + \frac{\text{Number of correctly paired retest-test samples}}{\text{Total number of participants}}}{2} \quad (2.7)$$

Algorithm 2 Matching Rate Computation

Input: $\mathbf{I} \in \mathbb{R}^{N \times N}$

```
1:  $\mathbf{I}_{mask} \leftarrow \mathbf{I}$ 
2:  $MR_1 \leftarrow 0$ 
3: for  $i \leftarrow 1$  to  $N$  do
4:    $[maxValPerCol, rowIndices] \leftarrow \max(\mathbf{I}_{mask}, [], 1)$ 
5:    $[\sim, colIndex] \leftarrow \max(maxValPerCol)$ 
6:   if  $(colIndex == rowIndices(colIndex))$  then
7:      $MR_1 \leftarrow MR_1 + 1$ 
8:   end if
9:    $\mathbf{I}_{mask}(:, colIndex) \leftarrow -\text{inf}$ 
10:   $\mathbf{I}_{mask}(rowIndices(colIndex), :) \leftarrow -\text{inf}$ 
11: end for
12:  $\mathbf{I}_{mask} \leftarrow \mathbf{I}^T$ 
13:  $MR_2 \leftarrow 0$ 
14: for  $i \leftarrow 1$  to  $N$  do
15:    $[maxValPerCol, rowIndices] \leftarrow \max(\mathbf{I}_{mask}, [], 1)$ 
16:    $[\sim, colIndex] \leftarrow \max(maxValPerCol)$ 
17:   if  $(colIndex == rowIndices(colIndex))$  then
18:      $MR_2 \leftarrow MR_2 + 1$ 
19:   end if
20:    $\mathbf{I}_{mask}(:, colIndex) \leftarrow -\text{inf}$ 
21:    $\mathbf{I}_{mask}(rowIndices(colIndex), :) \leftarrow -\text{inf}$ 
22: end for
23:  $Avg_{MR} \leftarrow \frac{MR_1 + MR_2}{2 \times N}$ 
```

Output: Avg_{MR}

2.2.9 Fingerprinting Framework Adapted to Tucker Decomposition

The proposed fingerprinting framework consists of five key steps: *i)* given a data acquisition session (either test or retest) of an fMRI condition, construct a tensor that contains all participants FCs, *ii)* decompose the tensor via Tucker decomposition to obtain a core tensor, a brain parcellation factor matrix, and a participant factor matrix, *iii)* estimate the other session’s participant factor matrix based on the decomposition of the given session, *iv)* obtain an identifiability matrix by computing pairwise Pearson’s correlation between the rows of both participant factor matrices, and *v)* calculate the matching rate for the obtained identifiability matrix. A schematic representation of our framework is presented in Figure [2.1](#).

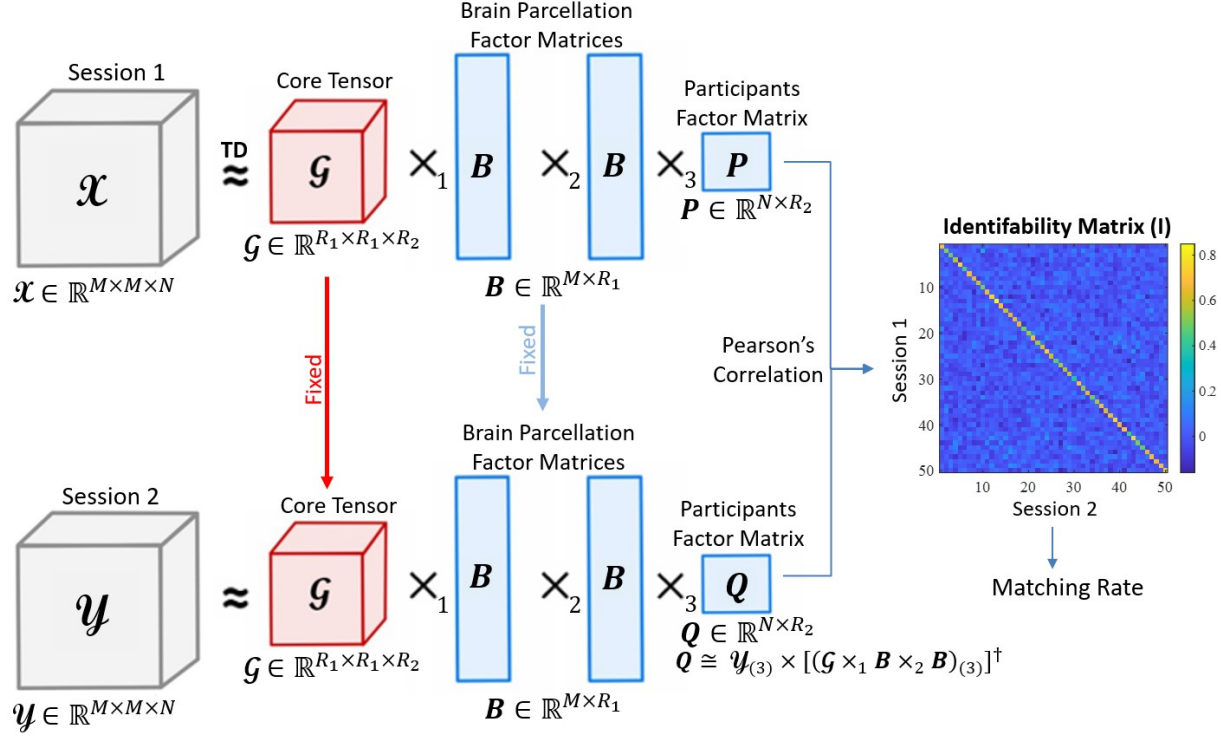


Figure 2.1. Graphic representation of the participant identifiability framework. For an input tensor $\mathcal{X}$ consisting of FCs from one imaging session, we compute its Tucker decomposition to obtain a core tensor $\mathcal{G}$, two identical brain parcellation factor matrices $\mathbf{B}$, and a participants' factor matrix $\mathbf{P}$. Fixing $\mathcal{G}$ and $\mathbf{B}$ factor matrices, we estimate the participant's factor matrix $\mathbf{Q}$ for the FCs from another imaging session $\mathcal{Y}$. We then obtain an identifiability matrix $\mathbf{I}$ by computing row-wise Pearson's correlation between $\mathbf{P}$ and $\mathbf{Q}$, and obtain a matching rate score through $\mathbf{I}$.

In the following section, we discuss how matching rate is affected by parcellation granularity, decomposition rank, scanning length of fMRI conditions, and under within- and between-condition scenarios.

2.3 Results

For all eight fMRI conditions, the proposed fingerprinting framework was applied to two main settings. First, when test and retest FCs correspond to the same fMRI condition (within-condition fingerprinting). Second, when combining resting-state FCs with task FCs (between-condition fingerprinting). For the aforementioned settings, we further investigated

the impact of brain parcellation rank and participant rank on Tucker decomposition and subsequent fingerprinting, and whether scanning length duration has a significant effect on matching rates.

2.3.1 Parcellation Granularity Effect on Fingerprinting

Finer parcellation granularities offer a more detailed representation of the functional connectome by dividing the brain into more regions. However, this comes with a trade-off, as the time series for each region may be less reliable due to being derived from a smaller number of voxels. Hence, FCs fingerprint are expected to be dependent on the parcellation granularity of the data. Figure 2 displays the within-condition fingerprinting results of maximum-rank ST-HOSVD decompositions (i.e, for parcellation 614, the brain parcellation rank R_1 and participant rank R_2 were chosen to be 614 and 426, respectively) for parcellation granularities of 114, 214, 314, 414, 514, 614, 714, 814, and 914. As the parcellation granularity increases, matching rates plateau for all fMRI conditions starting at parcellation granularity of 414. Therefore, we give a stronger emphasis to parcellation granularity of 414 in the remaining analyses of this chapter. It is noteworthy that the longest (resting-state) and shortest (Emotion) fMRI conditions display, respectively, the highest and lowest matching rate values across all parcellation granularities. However, tasks with intermediate scanning length do not follow such trend.

2.3.2 Dimensionality Reduction Methods in Fingerprinting

In Table 2.2, a comparison between different dimensionality reductions methods versus a baseline (consisting of vectorized FCs) for parcellation granularity of 414 is shown. When computing PCA, the number of principal components used to obtain the identifiability matrix was chosen to be the maximizer of subject identifiability as introduced in the work of Amico and Goñi (2018) [9]. The reported matching rates for CP decomposition are the highest among rank- R decompositions, with $R \in \{100, 200, 300, 400, 414\}$. Equivalently, the reported matching rates for Tucker based methods are the highest among all considered combinations of brain parcellation rank R_1 and participant rank R_2 . Among the presented

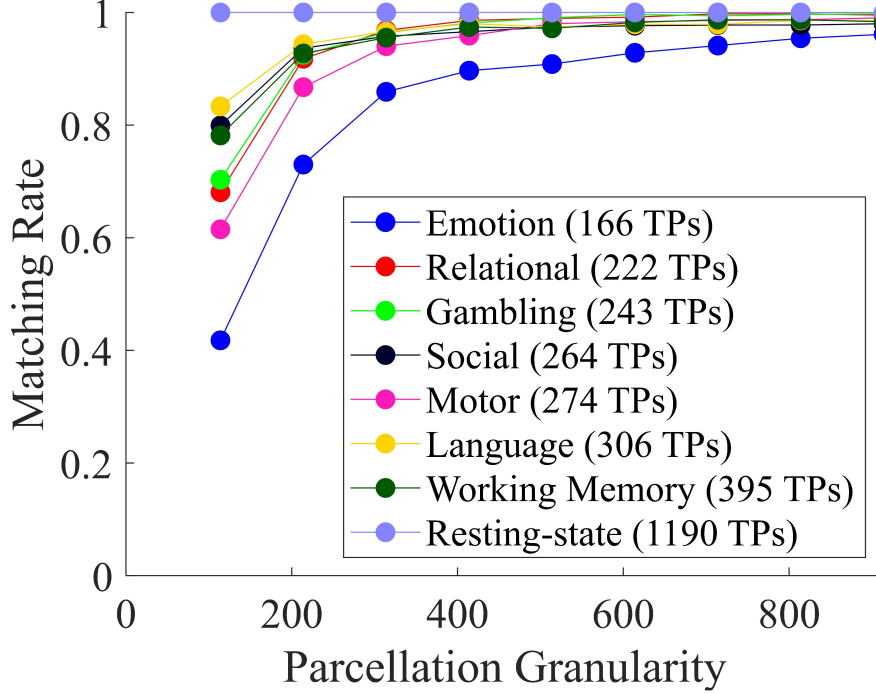


Figure 2.2. Fingerprinting gradient as measured by matching rate for all fMRI conditions. For each parcellation granularity, the brain parcellation rank and participant rank were set to their maximum value.

dimensionality reduction methods, Tucker decomposition (ST-HOSVD and HOOI) demonstrates, by a considerable margin, the highest fingerprint.

Considering the high-dimensionality of neuroimaging data, it is crucial to consider the scalability of Tucker-based methods for the purpose of fingerprinting. In Table 3, the runtime (seconds) of the most computationally demanding decomposition of a tensor of dimensions $PG \times PG \times 426$, where PG denotes the parcellation granularity, is presented for each dimensionality reduction method. For CP and Tucker-based methods, the highest runtime corresponds to a decomposition of maximum possible rank. For example, for a tensor of dimensions $414 \times 414 \times 426$, the CP rank is set to $R = 414$, and the ranks for ST-HOSVD and HOOI are set to $R_1 = R_2 = 414, R_3 = 426$. Equivalently, a matrix of dimensions $\binom{414}{2} \times 852$ (columns represent the upper triangular entries of vectorized FCs) all 852 principal component are computed for PCA. As the dimensionality of the data increases with the parcel-

Table 2.2. Matching rate comparison between (baseline) FCs and distinct dimensionality reduction methods for parcellation granularity of 414. Bold values highlight the highest matching rates among all techniques for each fMRI condition, which are presented in ascending order of scanning length (“TPs” represent the number of time points in the time series of each condition).

fMRI Conditions	FCs	PCA	CP	ST-HOSVD	HOOI
Emotion (166 TPs)	0.57	0.63	0.64	0.90	0.90
Relational (222 TPs)	0.79	0.84	0.90	0.99	0.99
Gambling (243 TPs)	0.84	0.86	0.93	0.98	0.98
Social (264 TPs)	0.88	0.88	0.92	0.97	0.97
Motor (274 TPs)	0.82	0.82	0.75	0.96	0.96
Language (306 TPs)	0.91	0.90	0.92	0.98	0.98
Working Memory (395 TPs)	0.86	0.84	0.92	0.97	0.97
Resting-state (1190 TPs)	0.92	0.97	0.99	1.00	1.00

lation granularity, PCA tends to become less computationally expensive than ST-HOSVD. However, the increase in computational cost is negligible compared to the effectiveness of ST-HOSVD in uncovering fingerprints. For reference, the reported runtimes were measured on a Lenovo Legion 5i laptop equipped with an Intel Core i7 processor, 32GB of RAM, and an NVIDIA GeForce RTX 3070 Ti GPU.

Table 2.3. Comparison between the worst runtimes of different dimensionality reduction methods. For each parcellation granularity PG , a tensor of dimension $PG \times PG \times 426$ was decomposed by CP Decomposition, ST-HOSVD, and HOOI. Equivalently, PCA was applied to a matrix of dimension $\binom{PG}{2} \times 852$, where the rows consist of vectorized upper triangular FCs, and the columns correspond to the participants’ test and retest sessions. Bold values indicate the shortest runtime across methods.

Method	114	214	314	414	514	614	714	814	914
Runtime (seconds) per Parcellation Granularity									
PCA	0.38	0.85	1.93	3.74	6.26	8.75	12.45	21.23	24.71
CP Decomposition	2.15	12.88	45.82	108.23	158.75	222.00	288.06	378.45	465.32
ST-HOSVD	0.38	0.73	1.58	3.20	5.15	8.78	13.24	19.55	27.62
HOOI	0.87	1.76	3.81	6.41	11.70	18.03	26.51	36.20	72.07

2.3.3 Evaluating the Impact of Brain Parcellation Rank and Participant Rank on Fingerprinting.

To obtain a holistic view of how brain parcellation rank and participant rank affect fingerprinting, we computed matching rates under different combinations of both ranks for all fMRI conditions. Results shown in Figure 2.3 indicate that higher brain parcellation ranks led to higher matching rates compared to lower brain parcellation ranks. However, the impact of the participant rank on matching rates depends on the fMRI condition. Specifically, with Emotion and Motor tasks we can achieve near-optimal matching rates with a participant rank of 300 or higher, while for all the other fMRI tasks we achieve near-optimal matching rates earlier, starting at participant rank of 150. Resting-state matching rates were the highest among all fMRI conditions, reaching optimal scores starting at participant rank 100, with 100% matching accuracy. In contrast, Emotion had the lowest matching rate.

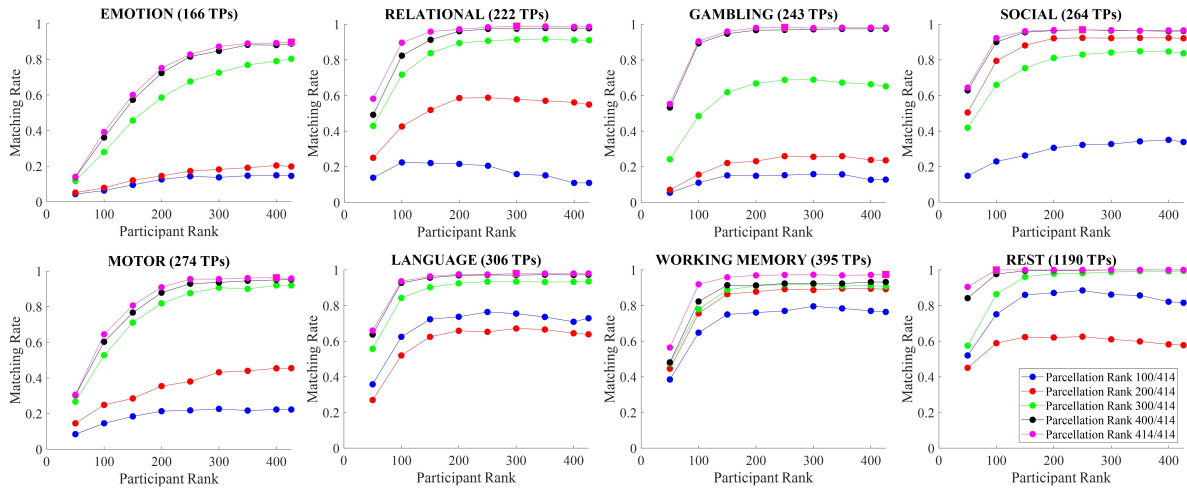


Figure 2.3. Brain parcellation rank and participant rank effects on matching rate for all eight fMRI conditions, ordered by scan length (represented by the number of time points or “TPs” in their time series) for granularity of Schaefer parcellation 414. For each condition, the fingerprinting framework is applied in both test→retest and retest→test directions. Each curve illustrates the average matching rate computed across both settings at each combination of brain parcellation ranks and participant ranks.

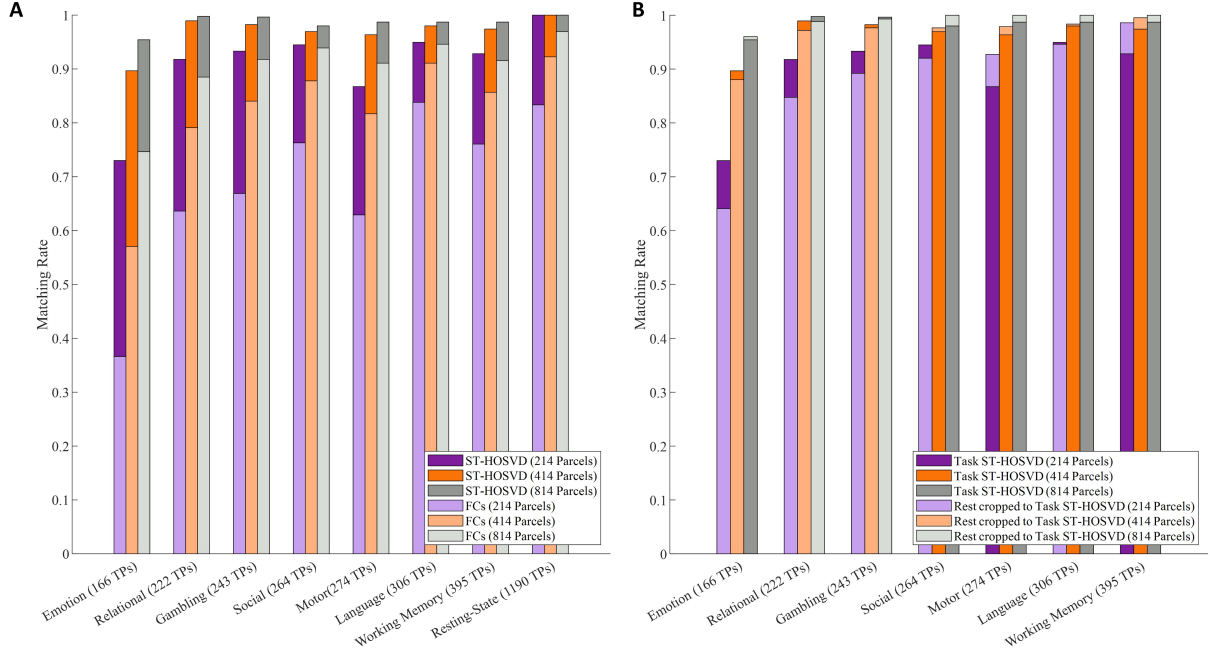
2.3.4 Within-Condition Fingerprinting

The concept of within-condition fingerprinting reflects the ability to correctly match two scans of the same participant when all evaluated scans belong to the same fMRI condition. The proposed fingerprinting framework yields a substantial matching rate increase across all considered parcellation granularities relative to vectorized original FCs, which were adopted as baseline. For original FCs, identifiability matrices were obtained by computing pairwise Pearson’s correlation between test and retest vectorized upper triangular original FCs.

In Figure 2.4A, it is shown, for each condition, the highest matching rate obtained for all possible combinations of brain parcellation rank and participant rank. The highest matching rate increase with respect to original FCs was observed for parcellation granularity of 214, for which the fingerprinting framework generated matching rate improvements ranging from 11% (Language) to 36% (Emotion). Resting-state matching rates were the highest in all parcellations, achieving 100% for all parcellation granularities.

Given the mismatch in scan duration between resting-state and tasks, a matching rate comparison between fMRI tasks and resting-state when resting-state time series are cropped to match the duration of tasks is shown in Figure 4B. To do so, and in order to avoid possible biases when using the first time-points, resting-state FCs are estimated by using the L central time points, where L is the duration of the corresponding task. For conditions with short scanning length (Emotion, Relational, and Gambling), resting-state displays a lower fingerprint than the fMRI tasks in all parcellation granularities. The previous is also true for Social and Language tasks and parcellation granularity 214.

The fingerprinting framework was carried out in two ways. First, by inputting a tensor consisting of test FCs while aiming to estimate the participants factor matrix of the retest session. Second, by inputting a tensor consisting of retest FCs while aiming to estimate the participants factor matrix of the test session. The results of both procedures were averaged for all combinations of ranks considered. The reported values correspond to the highest of such averages.



2.3.5 Between-Condition Fingerprinting

Between-condition fingerprinting measures the degree to which we can match, for each participant, a scan from one fMRI condition to their scan from another fMRI condition. Using our fingerprinting framework, we estimate the participant factor matrix from a tensor consisting of task (test) fMRI by inputting a tensor consisting of resting-state (retest) FCs. Given that the scanning length of resting-state is substantially longer than the length of all fMRI tasks, we consider two scenarios when estimating resting-state FCs: *i*) using the full resting-state time series and *ii*) matching (hence reducing) the number of timepoints in the resting-state time series to the duration of each task (e.g. when pairing with Emotion

task, resting-state FCs are computed using 166 out of the total 1190 time points). In the latter case, we further explore two strategies for sampling time points of the resting-state scan: *i*) randomly sample time points, and *ii*) randomly sample a starting time point in the range of $[1, 1190 - \text{length}(\text{task})]$ and take the randomly sampled starting time point and its $\text{length}(\text{task}) - 1$ consecutive time points.

Between-Condition Fingerprinting with Resting-state Full Scanning Length

The between-condition matching rates when resting-state FCs are estimated using the time series of the full scanning length are shown in Figure 2.5. Matching rates ranged from 76% (Relational) to 91% (Language). In Figure 2.5, we observe a high sensitivity to brain parcellation rank, with 414 being the optimal rank. Comparatively, matching rate also benefits from a higher participant rank. However, the degree to which this occurs is task dependent (i.e. a participant rank of 350 is nearly optimal for Emotion, but for Relational there is a clear benefit in going up to rank 426).

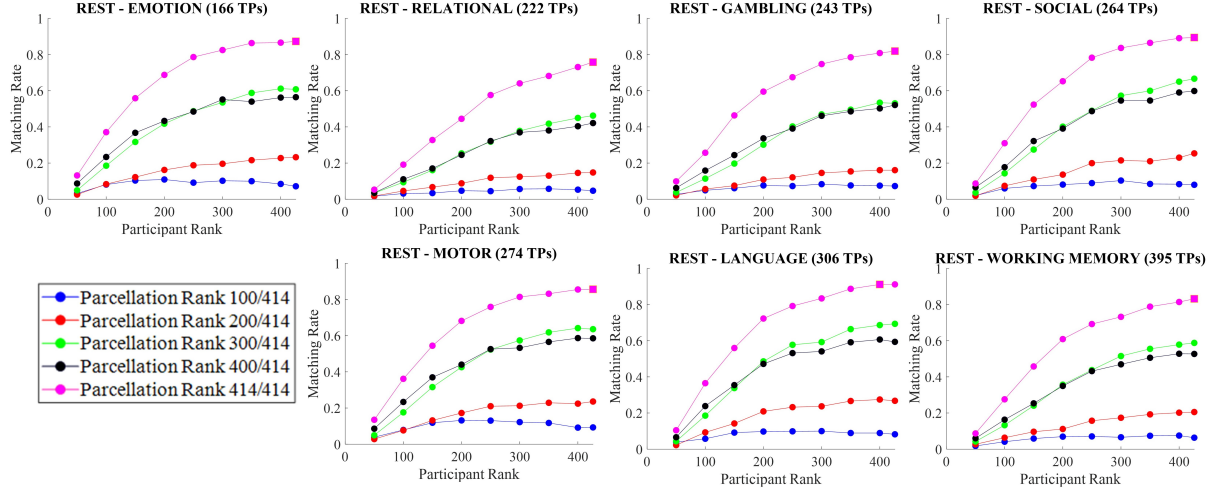


Figure 2.5. Parcellation rank and participant rank effects on between-condition matching rate for granularity of Schaefer parcellation 414. The fingerprinting framework was carried out by computing the ST-HOSVD of a tensor consisting of concatenated resting-state retest FCs (estimated using the entire scanning length of resting-state BOLD time series) from all participants. The obtained core tensor and brain parcellation factor matrix were then used to estimate the participants factor matrix of a tensor consisting of concatenated test FCs of fMRI tasks. A similar trend is observed in all plots, with parcellation ranks of 100 and 200 being insufficient to uncover fingerprints, while the small increase in parcellation rank from 400 to 414 results in a substantial improvement in matching rates.

In Figure 2.6, we observe a major increase in between-condition matching rates using Tucker decomposition when compared to original FCs. The procedure used for computing the identifiability matrix for original FCs in the between-condition setting was analogous to the one in the within-condition setting. Matching rate improvements obtained with Tucker ranged from 43% (Relational) to 72% (Language).

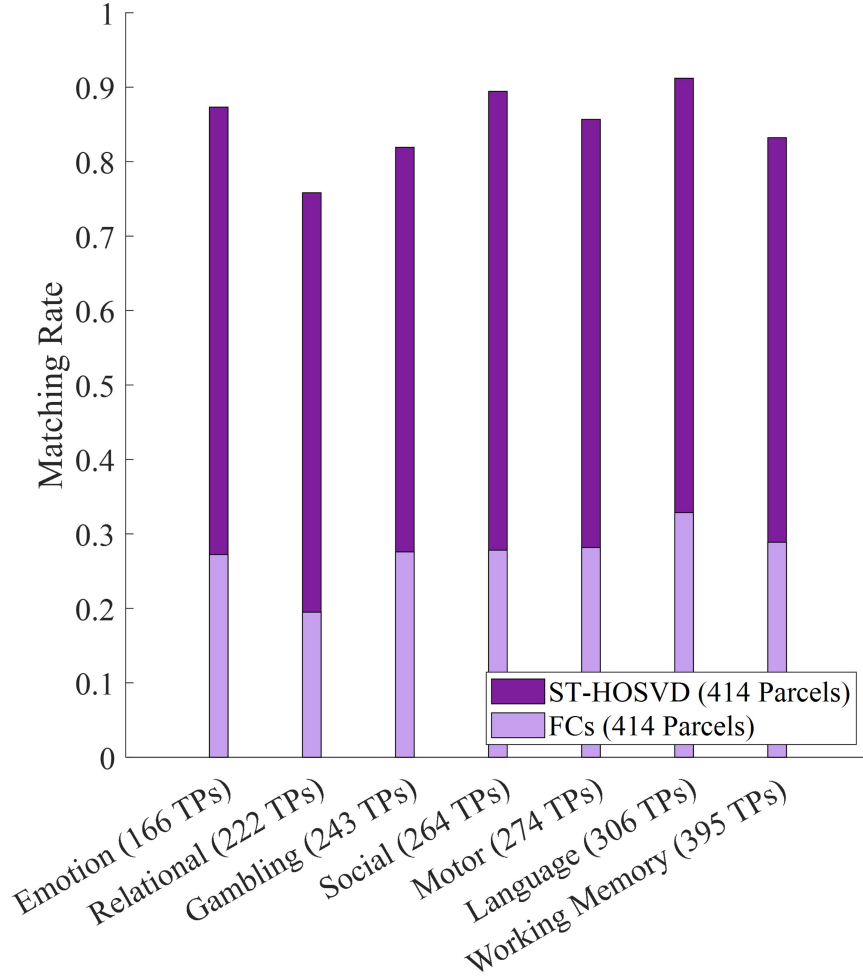


Figure 2.6. Between-condition comparison between highest matching rate obtained with the ST-HOSVD based fingerprinting framework and original FC matrices for parcellation granularity of 414. Despite being the shortest task and the one with worse within-condition fingerprints, Emotion displays one of the highest between-condition fingerprints. Contrarily, Working Memory possesses one of the highest fingerprints in the within-condition setting, but one of the lowest in the between-condition setting.

Between-Condition Fingerprinting with Resting-state Matched-to-Task Scanning-Length

In order to account for scanning length effects, the between-condition matching rate results were obtained when resting-state FCs are estimated by matching the time-series length to each fMRI task. Results are shown in Figure 2.7. For this analysis, we fixed the

brain parcellation rank to 414, which was the optimal choice in terms of fingerprinting as shown in Figure 2.5. For each participant rank across all between-condition settings, we computed matching rates with 100 different samplings of time points for the resting-state time series. In Figure 2.7A, we display the results of randomly sampling an initial time point and choosing the remaining time points to be consecutive to it. In Figure 2.7B, we display the results of randomly sampling all time points. When comparing both sampling strategies, randomly sampling all time points is a more robust and effective approach in terms of fingerprinting, as measured by the standard deviation and mean of each box plot.

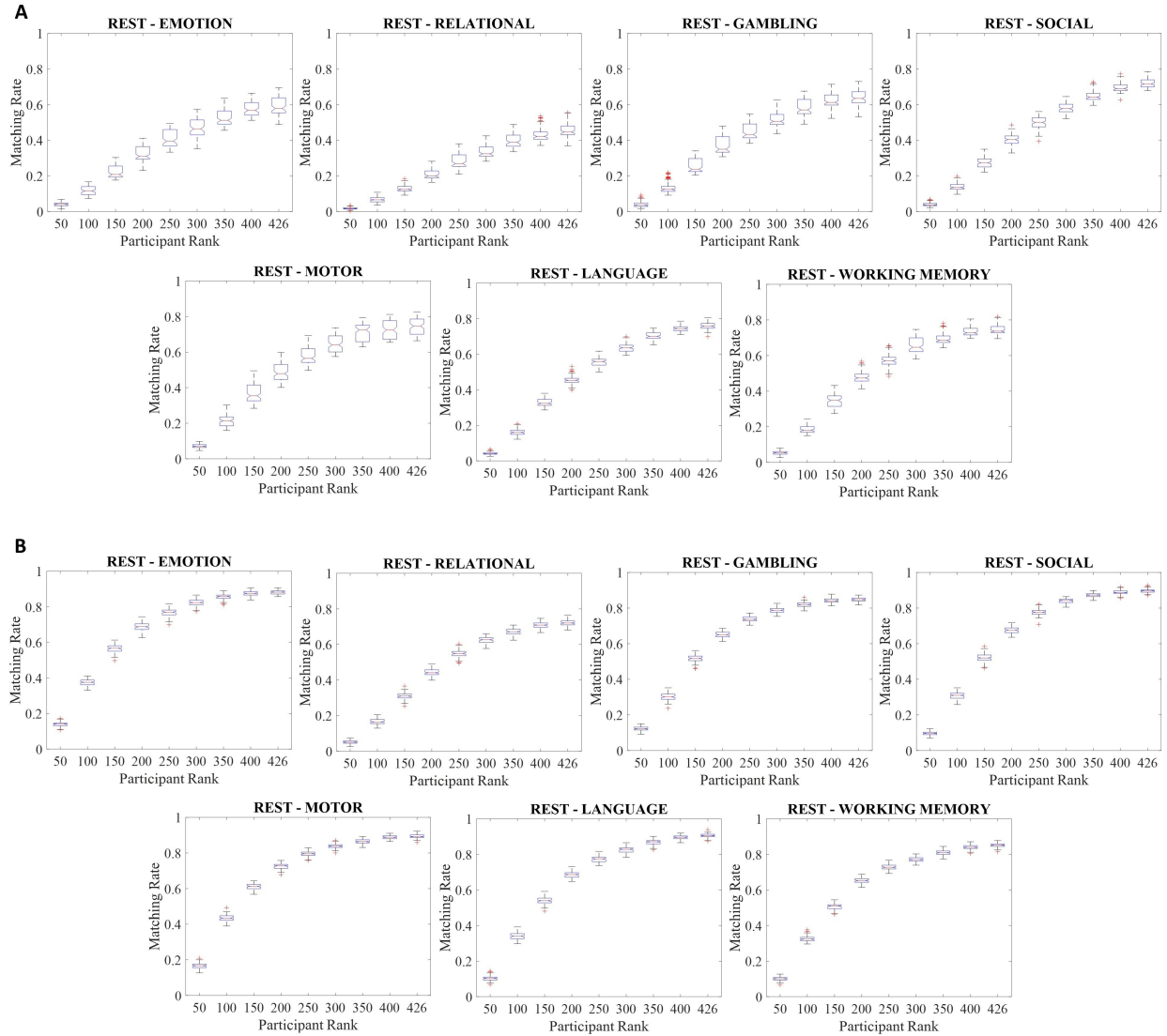


Figure 2.7. Between-condition matching rates obtained via ST-HOSVD when resting-state FCs are estimated using time series of equal length to those of each fMRI task for parcellation granularity of 414. The brain parcellation rank was set to 414, and 100 random sampling trials were conducted to select the time points used to estimate resting-state FCs for each participant rank. **A.** Rest (retest) - task (test) matching rate results when, starting with a randomly sampled time point, consecutive time points are used to construct resting-state FCs. In this scenario, matching rates grow at an approximately linear rate with participant rank. **B.** Matching rates when randomly sampled time points are used to construct resting-state FCs. Here, matching rates exhibit a bounded exponential pattern, reaching a plateau at participant rank 350.

2.3.6 Analysis of Misclassified Participants

Tucker decomposition led to a substantial improvement of matching rates for all fMRI conditions when compared to using FCs directly. Figure 2.8 provides additional evidence of how this is achieved. Figures 2.8A and 2.8B show, for Emotion and resting-state conditions and parcellation granularity of 414, the identifiability matrices obtained from FCs and ST-HOSVD along with the within- and between-participant distributions. When compared to results on FCs, similarities derived from ST-HOSVD tend to be diminished between participants (centered around 0), while remaining nearly the same within participants. In 2.8C, 2.8D, and 2.8E, the following cases are shown: in 2.8C, a participant whose Emotion test and retest FCs are incorrectly matched using both vectorized FCs and ST-HOSVD; in 2.8D, a participant whose Emotion test and retest FCs are incorrectly matched using vectorized FCs but correctly matched using ST-HOSVD; and in 2.8E, a participant whose resting-state test and retest FCs are incorrectly matched using vectorized FCs but correctly matched using ST-HOSVD.

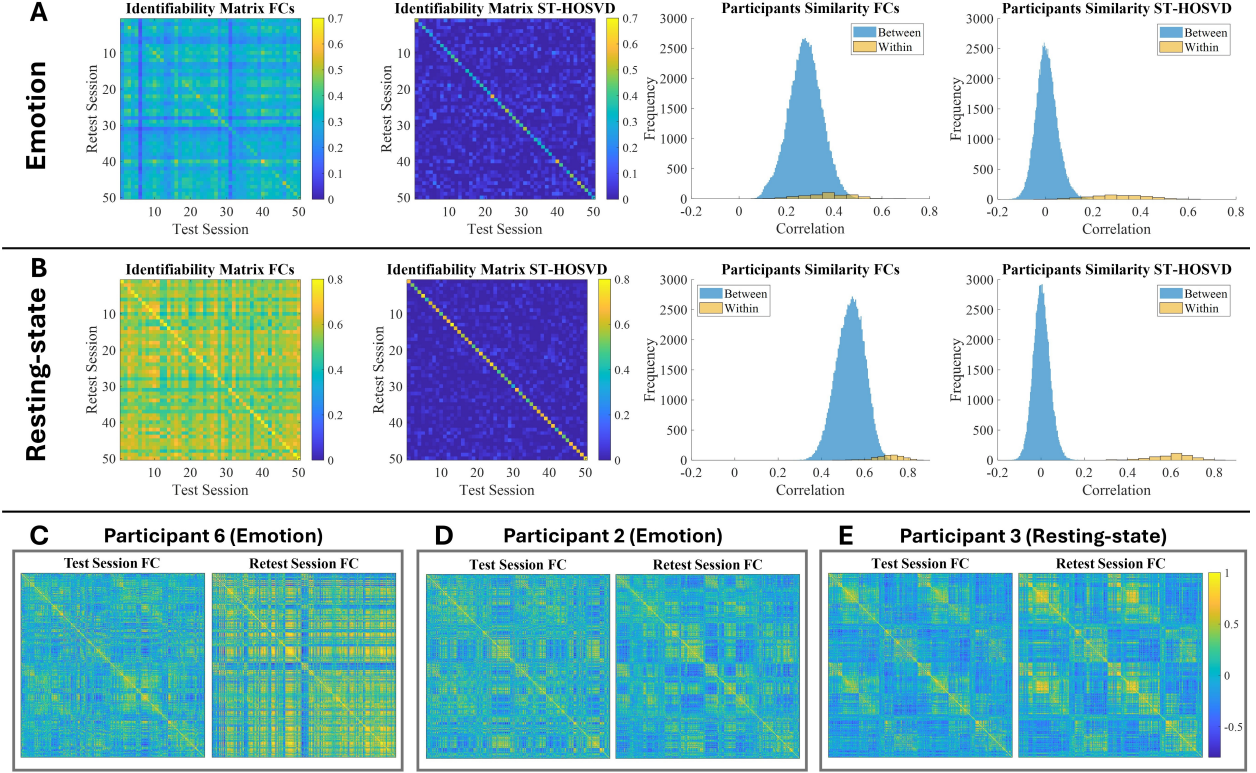


Figure 2.8. Comparison between tasks that possess the lowest (Emotion) and highest (Resting-state) within-condition fingerprints for parcellation granularity of 414. **A.** From left-to-right: first 50×50 block of the 426×426 identifiability matrix derived from vectorized FCs, first 50×50 block of the 426×426 identifiability matrix derived from the proposed fingerprinting framework based on ST-HOSVD, distribution of within-participant (main diagonal entries) and between-participant (off-diagonal entries) sessions of the FCs identifiability matrix, distribution of within-participant and between-participant sessions of the ST-HOSVD identifiability matrix. **B.** Analogous resting-state identifiability matrices and distributions. **C.** Participant whose Emotion test and retest FCs are incorrectly matched when computing the identifiability matrix via vectorized FCs and ST-HOSVD. **D** and **E.** Participants whose Emotion and resting-state FCs are incorrectly matched when computing the identifiability matrix using vectorized FCs, but correctly matched when computing the identifiability matrix via ST-HOSVD.

2.4 Discussion

Table 2.2 shows that under all conditions Tucker decomposition-based methods (ST-HOSVD and HOOI) consistently achieve the highest matching rates compared to FCs and other data-driven techniques. Both methods significantly outperform PCA and CP, particularly in conditions with fewer time points, such as Emotion and Relational, where FCs, whether used directly or in combination with PCA under-perform. The results highlight the effectiveness of Tucker decomposition techniques in enhancing fingerprinting accuracy even for fMRI conditions with a short scanning length.

The results presented in Figure 2.2 provide meaningful insights into the influence of parcellation granularity on fingerprinting accuracy. The observed matching rate plateau at higher granularities indicates that beyond a certain threshold, increasing the parcellation resolution offers diminishing matching rate returns, which could be due to an increase in noise in the overall data of each participant. This is consistent with prior findings by Finn et al. (2015) [7], who showed that measures from FCs estimated with long scan sessions provide meaningful information about individuals even with moderate parcellation granularity. The significantly higher matching rates for resting-state compared to tasks aligns with previous research showing that resting-state data captures more stable and individualized connectivity patterns due to its longer scanning length and more consistent brain-wide activity [9]. However, it has also been shown that when accounting for scanning length, resting-state has a lower fingerprint than tasks [13]. Interestingly, the lower performance observed for Emotion reflects the challenges of identifying individuals from shorter or more transient brain states, where brief tasks led to less reliable fingerprinting. The lack of a strict linear relationship between task length and matching rates across intermediate conditions suggests that certain cognitive tasks elicit more distinct and stable connectivity patterns, regardless of their duration. This nuanced relationship highlights the complexity of functional connectome dynamics, emphasizing the need for both sufficient data length and appropriate parcellation resolution to maximize fingerprinting reliability.

As shown in Figure 2.3, compressing the brain parcellation information is detrimental to fingerprinting, as the highest matching rates were obtained with a brain parcellation rank of

414. In contrast, compressing the participants specific dimension overall preserves matching rate, as shown by the high matching rates obtained with participant rank as low as 150 for Relational, Gambling, Social, Language, Working Memory, and Resting-State, and rank of 300 for Emotion and Motor. The results from this analysis imply that the dimension of the participants information, represented by the participants rank, can be considerably compressed while preserving fingerprints. This indicates that there is some redundancy in inter-participant variability, with not every participant adding new information to the data. However, it is important to emphasize that the compressibility of the participants dimension is likely cohort-size dependent, and must therefore be reexamined when dealing with different cohort sizes.

The within-condition results shown in Figure 2.4A confirm the presence of functional connectome fingerprints and demonstrate that the proposed framework is particularly effective in uncovering them, compared to non-decomposed functional connectomes. Notably, the obtained fingerprinting results were substantially higher than those obtained with FCs, especially for parcellation granularity 214, where fingerprinting performance was notably low (ranging from 37% to 83%). Noting that resting-state achieves matching rates of 100% in Figure 2.4A, in Figure 2.4B a comparison between tasks and resting-state is done when the FCs of both are computed with an equal number of time-points. The higher matching rates obtained from Emotion, Relational, and Gambling in comparison to resting-state suggesting that, to some degree, the high fingerprints derived from resting-state stem from the considerably longer duration of resting-state relative to fMRI tasks.

Unlike during fMRI tasks, participants in the resting-state condition are not engaged with any specific stimulus and therefore resting-state FCs encode “baseline” functional couplings among brain regions. Therefore, between-condition analyses allow us to assess the extent to which these couplings captured by the resting-state brain parcellation factor matrix can be recovered when participants engage in tasks. Even though extracting fingerprints in this setting is inherently more challenging compared to the within-condition setting, we were able to substantially increase matching rates relative to using original FCs. In Figure 2.5, it is shown that results are more sensitive to the parcellation and participant ranks, with a clear benefit in setting both to be the maximum value possible in many of the tasks. Further-

more, in Figure 2.6 it is shown that ST-HOSVD provided an even greater improvement over matching rates relative to FCs compared to the within-condition fingerprints. This supports the rationale that baseline functional couplings exist and can also be effectively uncovered while participants are engaged in a task.

We also explored the effect of reducing the resting-state BOLD time series duration to match the duration of each task. Doing so enables determining whether the entire scanning length is necessary to extract “key features” that enable obtaining between-condition fingerprints. From both sampling procedures carried out in this chapter, it is clear that randomly sampling time points is the superior strategy for fingerprinting. Comparing the results of 2.7A and 2.7B, we see that randomly sampling time points of resting-state scans is not only more effective than sampling them consecutively, but also as effective as constructing resting-state FCs using the full scan length (as shown in Figure 2.5). This result aligns with the fingerprinting improvements seen when sub-sampling frames of edge-based time-series [46].

The results presented in Figure 2.8 highlight the effectiveness of the proposed ST-HOSVD-based framework in enhancing FC fingerprinting relative to using FCs directly. Panels **A** and **B** illustrate the identifiability matrices and participant similarity distributions for Emotion and resting-state, respectively. The ST-HOSVD framework considerably improves the separation between within-participant and between-participant similarity, as evident by the clearer diagonal structure in the identifiability matrices and the sharper peak of the within-participant similarity distribution. This indicates that the participants factor matrix derived from ST-HOSVD better captures the individual-specific features of FC patterns compared to vectorized FCs. Panels **C**, **D**, and **E** respectively display qualitative examples in which the proposed framework fails to correctly match the Emotion FCs of one participant, but succeeds in matching the FCs of participants that cannot be matching using FCs directly. Visually speaking, it is easy to see similarities between the FCs from participants 2 and 3, however the same cannot be said about the FCs from participant 6. While these results demonstrate the improved fingerprinting capability of the ST-HOSVD-based framework, potential limitations exist. Factors such as variations in FC stability across tasks or inter-individual differences in connectivity patterns could influence the framework’s

performance. Additionally, differences in cohort size, scanner parameters, or preprocessing pipelines may affect the generalizability of the results, and should be further analyzed in future studies.

Both CP and Tucker decomposition are commonly used tensor decomposition techniques for dimensionality reduction and feature extraction purposes. In the context of fingerprinting, CP falls short due to two key reasons. First, the assumption that the original high-dimensional data can be reconstructed using non-interacting components is too restrictive, as we know that there are innate interactions between brain regions under a functional connectivity standpoint [47]. Second, due to CP decomposition being a single rank decomposition, we cannot freely explore the dynamics of compressing the different dimensions of the data. Conversely, the Tucker’s core plays a pivotal role in capturing the interactions between components while giving us the flexibility to explore how different levels of compression of the brain parcellation and participants information affect fingerprinting, thus overcoming both drawbacks of CP decomposition. However, drawing neuroscientific insights directly from the factor matrices derived from Tucker decomposition of FCs is non-trivial due to the existence of the core tensor [21], which captures several interactions between the components from each factor matrix.

FC fingerprinting, while having the potential to provide valuable insights in clinical and forensic settings, raises significant ethical concerns regarding privacy, bias, and potential misuse. The sensitive nature of neuroimaging data, which can reveal information about an individual’s cognitive state, mental health, or even predispositions to certain conditions, makes it highly vulnerable to privacy breaches. Without proper anonymization, such data could be exploited for unauthorized profiling or discrimination. To safeguard privacy, robust anonymization techniques, such as de-identification and differential privacy, should be implemented. Additionally, data security measures, including encryption during storage and transmission, as well as strict access controls, are essential to prevent unauthorized usage. Bias is another concern, as models trained on non-representative datasets may lead to inaccurate or unfair identifications, particularly in forensic applications. Ensuring diverse, unbiased datasets and regularly auditing algorithms for fairness can help mitigate this risk. Address-

ing these ethical implications is crucial to prevent the misuse of neuroimaging fingerprinting and protect individual rights.

Our study has limitations. As discussed above in detail, interpreting the factor matrices derived from Tucker decomposition is not straightforward due to the presence of a core tensor that captures complex interactions between all factor matrices [21]. Additionally, the proposed fingerprinting framework does not allow for incremental updates to the core and factor matrices when FCs from new participants are introduced. Rather, the entire Tucker decomposition and fingerprinting framework would need to be recomputed. Our study leads to several avenues for further work. When preprocessing fMRI BOLD data, there is a large number of pipelines, steps, and parameters that can be used, with each specific configuration possibly leading to different FC estimations and ultimately differences in fingerprinting. Further work could assess the specific impact of such decisions (e.g. global signal regression, bandpass filter) on the association between Tucker decomposition and matching rates. Also, while Pearson’s correlation is the most widely used coupling method for fMRI time-series to estimate functional connectivity, other alternatives such as mutual information should be considered in order to assess the impact of different coupling methodologies when using decomposition methods to assess fingerprinting. To improve the interpretability of Tucker decomposition, future work could explore strategies to extract meaningful neuroscientific insights from the core tensor and factor matrices. One promising approach is to impose sparsity constraints on the core tensor using L_0 or L_1 regularization, which could help isolate dominant functional connectivity patterns shared across individuals. Simultaneously, the participant factor matrix would reveal the contribution of each underlying pattern to an individual’s FC, thereby enhancing interpretability. Another possible path for deriving neuroscientific insights is to perform post-hoc statistical analyses by correlating the components from the factor matrices with cognitive or behavioral measures. Doing so would help bridge the gap between Tucker-based decompositions methods and neuroscientific interpretation, allowing for a better understanding of how extracted patterns relate to individual differences in cognition, behavior, or clinical state.

2.5 Conclusion

In this chapter, a mathematical framework based on Tucker decomposition is presented for functional connectome fingerprinting. The ability to project high-dimensional functional connectivity data to lower dimensional spaces enables separately extracting cohort-level functional patterns, encoded by the brain parcellation factor matrix, and participant-specific patterns, contained in the participants factor matrix. Through within- and between-condition analyses, it is shown that the participant factor matrix serves, to a great extent, as a fingerprint of the participants. This is supported by the obtained matching rates ranging from 90 – 100% in the within-condition setting, which are consistently higher than those obtained from FCs directly or PCA, and 76 – 92% in the between-condition setting for the intermediate parcellation granularity of 414. Furthermore, the results from this chapter suggest that fingerprinting of rest vs task is highly dependent on scanning length (see Figure 2.4B). For up to 275 TPs (3 minutes 18 seconds), resting-state FCs had a smaller matching rate than their corresponding tasks FCs (Emotion, Relational, Gambling, Social). Beyond 275 TPs resting-state FCs showed a higher matching rate than their corresponding task FCs (Motor, Language, Working Memory). Overall, tensor-based methods demonstrate a high potential to uncover functional connectivity fingerprints. Our work has implications in understanding how data-driven decomposition methods enable us to assess fingerprinting with respect to: i) brain parcellation granularity from coarse grain (114) to fine grain parcellations (914); ii) dimensionality reduction for brain parcellation rank and for participants rank; iii) scanning length of the fMRI condition; iv) between and within fMRI condition.

3. ROBUSTNESS OF TANGENT FCS’ FINGERPRINTS

3.1 Introduction

The human brain is one of the most complex biological systems known, exhibiting intricate structural and functional connectivity patterns that underpin cognition, behavior, and neurological disorders. The development of non-invasive neuroimaging techniques, particularly functional magnetic resonance imaging (fMRI), has revolutionized the ability to investigate functional brain organization [48] and its variability across individuals. In addition, neuroimaging techniques have contributed to the understanding of psychiatric disorders [49, 50], aging [51–53], and neuroplasticity [54–56]. Despite such advancements, obtaining high-quality, reproducible functional neuroimaging data remains an ongoing challenge. The reason for such arises primarily from two factors: (*i*) the substantial financial and infrastructural costs associated with fMRI experiments, restricting large-scale data acquisition to well-funded research institutions, and (*ii*) the inherent variability in fMRI experimental designs, which complicates the generalizability of findings across studies and populations [57].

Over the last decade, efforts related functional connectivity fingerprinting [7] have sought to demonstrate the consistency and reproducibility of individualized functional connectivity patterns. The concept of fingerprinting in neuroimaging has gained significant traction, as studies have demonstrated that an individual’s FC profile remains stable across different scanning sessions and conditions [9, 10]. However, reliably capturing this uniqueness requires robust mathematical frameworks capable of mitigating noise, improving inter-session reliability, and enhancing subject identifiability. Notably, recent studies have explored the use of Riemannian geometry-based methods, particularly tangent space projection, to improve the characterization of FC fingerprints [12, 13, 24].

Tangent space projection is a mathematical technique rooted in Riemannian geometry that provides a more stable representation of FC matrices by mapping them onto a Euclidean space [14]. This is particularly advantageous given that FC matrices reside on a non-Euclidean manifold of symmetric positive definite (SPD) matrices, where conventional statistical operations may not be optimal [58]. By leveraging the geometric properties of the

manifold, tangent space projection enhances the robustness of FC patterns across sessions, making it well-suited for fingerprinting applications. Recent studies have shown that using tangent space representations improves FC distinctiveness, thus enabling distinguishing classification performance in brain connectivity analyses [13, 24].

Given the importance of high-quality, reproducible neuroimaging data in functional connectivity studies, the Midnight Scan Club (MSC) dataset emerges as a unique and valuable resource. The MSC dataset comprises several hours of resting-state fMRI data collected from ten individuals across multiple scanning sessions, offering an unprecedented level of within-subject homogeneity [59]. This controlled acquisition setting provides the ideal conditions for assessing the reproducibility of FC patterns and evaluating the effectiveness of tangent space projection in fingerprinting applications.

In this study, we aim to:

1. Assess whether tangent FCs provide high distinctiveness across participants over multiple scanning sessions.
2. Investigate whether Euclidean metrics applied to tangent FCs enable recovering of corrupted information from participants under specific levels of regularization.
3. Determine whether sets of tangent FCs serve as effective objects for unsupervised learning tasks in neuroimaging.

By addressing these objectives, we seek to advance the application of Riemannian geometry in neuroimaging and contribute to improving the reliability of functional connectome fingerprinting methods.

3.2 Materials and Methods

In this section, a description of the MSC dataset is provided, as well as an overview of Riemannian geometry, tangent space projection of FCs, and t-stochastic neighbor embedding (t-SNE).

The dataset used in this study is derived from the Midnight Scan Club (MSC) dataset [59], a publicly available neuroimaging dataset designed to investigate the stability and

variability of individual functional connectivity (FC) patterns across multiple scanning sessions. The MSC dataset consists of high-resolution resting-state and task-based fMRI data collected from ten healthy adults across ten separate scanning sessions. By repeatedly acquiring data from the same individuals under controlled conditions, the MSC dataset offers a unique opportunity to assess the reproducibility of functional connectivity patterns, making it particularly well-suited for a study of functional connectome fingerprints.

The MSC dataset includes ten participants, as detailed in Table 3.1. Each participant underwent ten separate scanning sessions, each lasting approximately two hours, for a total of twenty hours of fMRI data per individual. The data comprises of four fMRI conditions: resting-state, Glasslexical, Memory, and Motor. A total of five hours of resting-state data and approximately five and half hours of task fMRI (combined between the three tasks) was collected for each participant. All scanning sessions were conducted at night to minimize the influence of daily cognitive fluctuations, hence the name of the dataset.

The extensive within-subject sampling across multiple sessions enables reliably analyzing the intra-individual variability in functional connectivity. Hence, the design ensures that any observed differences in FC patterns result from individual-specific features rather than session-to-session variability. The preprocessing procedure was done with the same pipeline as the one described in Chapter 1 for the HCP data, but with different band pass parameters.

Table 3.1. MSC subjects’ demographic information.

Participant	Gender	Age	Education Degree	Education Years
sub-MSC01	M	34	Doctorate	22
sub-MSC02	M	34	Doctorate	28
sub-MSC03	F	29	Masters	18
sub-MSC04	F	28	Bachelors	22
sub-MSC05	M	27	Bachelors	20
sub-MSC06	F	24	Bachelors	17.5
sub-MSC07	F	31	Masters	20
sub-MSC08	F	27	Professional	21
sub-MSC09	M	26	Professional	19
sub-MSC10	M	31	Professional	19

3.2.1 Riemmanian Geometry and Tangent Space Projection

Functional connectivity matrices estimated using pairwise Pearson correlation of blood oxygen level-dependent (BOLD) time series belong to the space of Symmetric Positive Definite (SPD) matrices, which forms a Riemannian manifold rather than a conventional Euclidean space. By definition, a Riemmanian manifold is a smooth manifold equipped with a Riemmanian metric, which defines an inner product for each tangent space of the manifold. Such manifolds are curved geometrical spaces, therefore the shortest path between two points, unlike in traditional Euclidean spaces, is not represented simply by a straight line, but rather by a curve that traverses the manifold's surface. Consequently, the use of metrics defined on Euclidean spaces to compare FCs is ill-advised as they can lead to inaccurate or misleading results by failing to capture the true geometric structure of FCs. Since FC matrices lie on Riemmanian manifolds, techniques from Riemmanian geometry such as tangent space projections provide mathematically sound tools to analyze and compare FCs [14].

Tangent space projection enables mapping FC matrices from the Riemmanian manifold to a locally Euclidean space, which intersects the manifold at a single reference point C_{ref} , which is typically given by a mean of FC matrices. This transformation allows FCs to be analyzed using standard Euclidean tools while preserving properties of the original space in which they reside. Mathematically, the tangent space projection $\vec{Q}$ of an FC Q is given by:

$$\vec{Q} = \log_m(C_{ref}^{-1/2} Q C_{ref}^{-1/2})$$

where $\log_m(\cdot)$ denotes the matrix logarithm. The tangent space project $\vec{Q}$ retains the same dimensions as Q , and thus can be used directly for fingerprinting analyses. In Figure 3.1, a schematic representation of tangent space projection is shown.

Tangent Space Projection

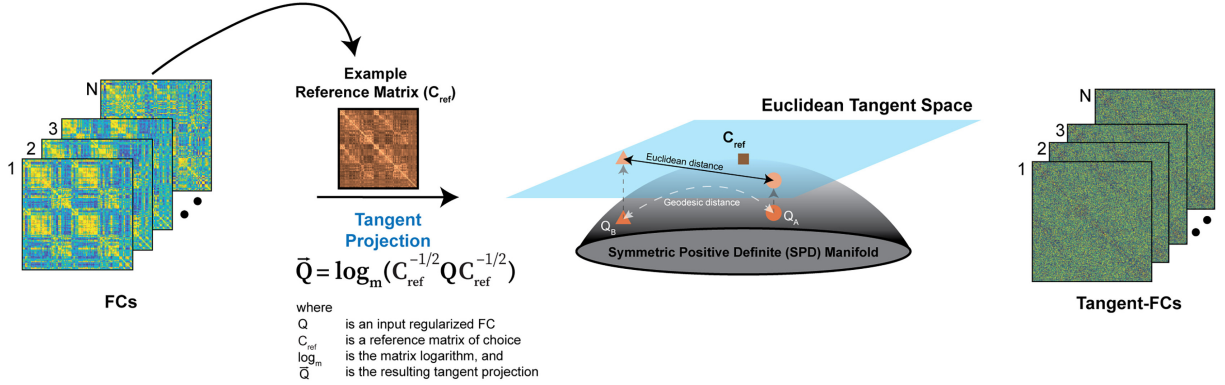


Figure 3.1. A representation of the tangent space projection of FCs. Based on a reference matrix C_{ref} lying on the manifold, a tangent space to which other FCs are projected is created. The resulting tangent FCs can be appropriately compared and studied with the use of Euclidean metrics. Image retrieved from Abbas et. al (2022) [13].

Given the exceptional fingerprints carried by tangent FCs [13] and the existence of optimal levels of FC regularization for FC fingerprinting [12], extend this approach to investigate its generalizability across multiple data acquisition sessions and different regularization values. Specifically, for a fixed fMRI condition, a meta-identifiability matrix is constructed to quantify subject-specific FC patterns.

The meta-identifiability matrix is computed as follows:

1. Vectorized tangent FCs are obtained for each participant across all sessions
2. Pearson's correlation and Euclidean distance are computed between every pair of vectorized tangent FCs.
3. Given that each participant undergoes 10 scanning sessions, the resulting meta-identifiability matrix has dimensions $(10 \text{ Participants} \times 10 \text{ Sessions}) \times (10 \text{ Participants} \times 10 \text{ Sessions}) = 100 \times 100$.

By evaluating the structure of this matrix, it is possible to assess the reliability of tangent FCs for individual fingerprinting in functional connectivity analysis.

3.2.2 t-Stochastic Neighbor Embedding of Tangent FCs

Functional connectivity data is inherently high-dimensional. This high dimensionality poses a challenge for interpretability and visualization of the data, making it difficult to directly observe individual differences or distinguish participants in an unsupervised fashion. To address this, we employ t-Distributed Stochastic Neighbor Embedding (t-SNE) [60], a widely used nonlinear dimensionality reduction technique that maps high-dimensional data into a lower-dimensional space, typically two or three dimensions. Unlike linear techniques such as Principal Component Analysis (PCA), which aims to decompose the data into components that explain most of its variance, t-SNE prioritizes the retention of local relationships between data points, making it particularly effective for visualizing high-dimensional data.

The core mechanism of t-SNE involves computing similarities between data points in both high-dimensional and low-dimensional spaces. In the high-dimensional space, pairwise distances between data points are converted into conditional probabilities, where similar points are assigned higher probability values. Let x_i and x_j be two high-dimensional data points. The similarity between such points is measured as the probability that x_j would be chosen as a neighbor of x_i under a Gaussian distribution N centered at x_i :

$$p_{ij} = \frac{\exp(-\|x_i - x_j\|^2/2\sigma^2)}{\sum_{k \neq l} \exp(-\|x_k - x_l\|^2/2\sigma^2)}$$

These probabilities are then mapped to a low-dimensional space, where a Student t-distribution with one degree of freedom is used to convert distances into probabilities. The joint probabilities q_{ij} are defined as:

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}}$$

where y_i and y_j are the low-dimensional representations of x_i and x_j . The use of Student's t-distribution mitigates the crowding problem that can emerge with traditional SNE.

Lastly, t-SNE minimizes the Kullback-Leibler divergence $\mathcal{L} = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$ between the high and low-dimensional distributions via gradient descent. For a meta identifiability matrix

constructed from vectorized tangent FCs, we apply t-SNE to uncover natural groupings of participants based on their functional connectivity patterns. Doing so enables us to assess whether tangent FCs preserve individual-specific fingerprints across multiple sessions by examining the extent to which FC representations from the same participant remain clustered together, hence providing an intuitive visualization of subject-level uniqueness.

3.3 Results

3.3.1 Original vs Tangent Meta Identifiability Matrices

To evaluate the effectiveness of tangent FCs in highlighting the distinguishability of subjects, we compare the original and tangent-space meta-identifiability matrices computed using Pearson’s correlation (PC) and Euclidean distance (ED), as shown in Figure 3.2. The matrices are structured such that both rows and columns correspond to subject-session pairings. For instance, the first 10 rows and columns represent the pairwise distance between all 10 vectorized FCs from subject 1.

The reference matrix C_{ref} was chosen as the mean FC from the first data acquisition session across all subjects. The results demonstrate that the tangent-space projection substantially enhances subject-specific distinctiveness. This effect is particularly noticeable when Pearson’s correlation is used as the distance measure, where the diagonal blocks representing intra-subject similarity across sessions become more prominent compared to off-diagonal elements. Additionally, the tangent-space projection aids in recovering corrupted sessions from Subject 8, reinforcing its robustness in mitigating noise and preserving individual-specific FC patterns. These findings highlight the utility of Riemannian geometry in improving the reliability of FC fingerprinting across multiple scanning sessions.

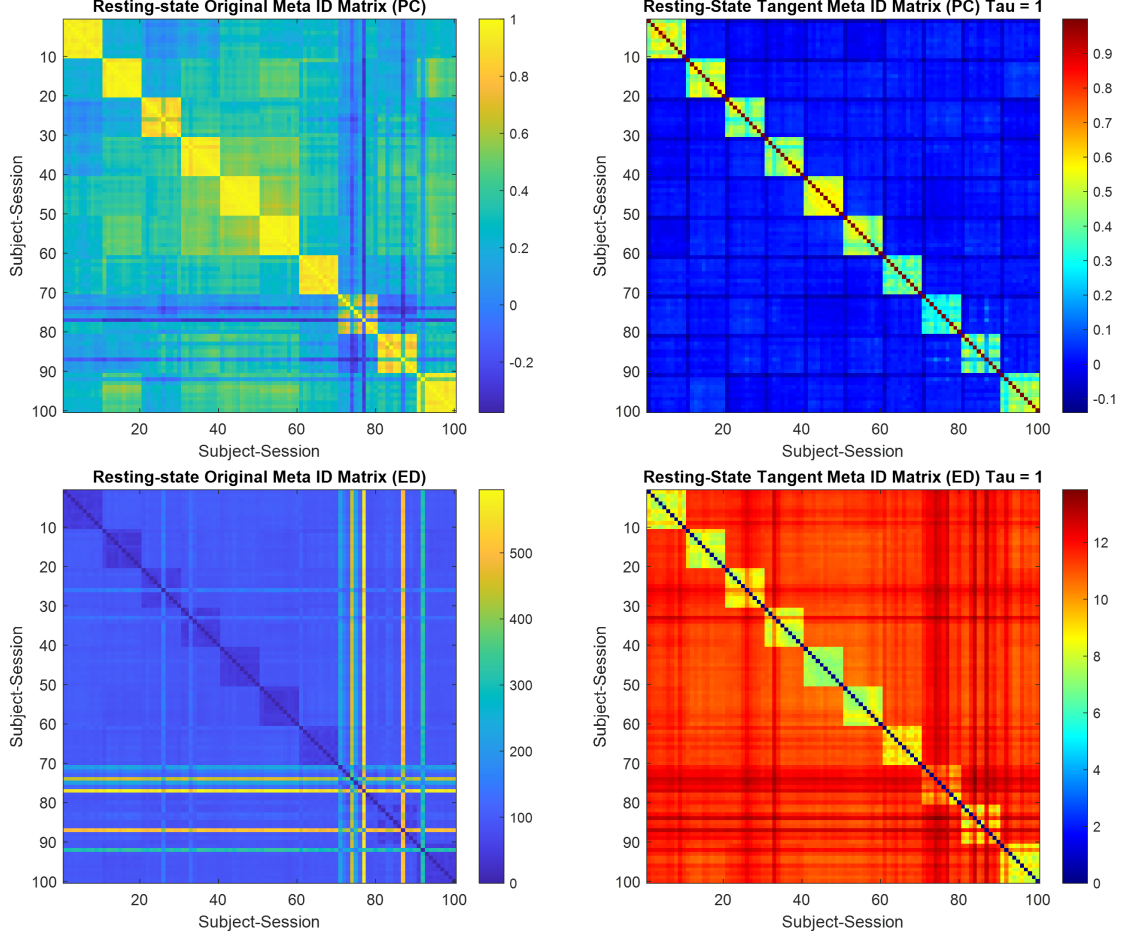


Figure 3.2. Comparison between original and tangent resting-state meta identifiability matrices. First row: comparison between original and tangent meta identifiability matrices when Pearson’s correlation is used as a distance measure between vectorized FCs. Second row: comparison between original and tangent meta identifiability matrices when Euclidean distance is used as a distance measure between vectorized FCs. Regularization values $\tau = 1$ was used for both tangent-space projections.

3.3.2 t-SNE of Meta Identifiability Matrices

t-SNE was applied to the meta identifiability matrices obtained from both original and tangent FCs. As figure 3.2 demonstrates higher subject distinctiveness when Pearson’s correlation is used rather than Euclidean distance, the analysis of t-SNE was restricted to the meta identifiability computed with Pearson’s correlation. For the purpose of applying t-SNE, each row of the meta identifiability matrices corresponds to an observation, with a number

of features equal to the total number of subject sessions ($\#$ Total Subject Sessions = 100). Figure 3.2 illustrates the results of applying t-SNE to both meta-identifiability matrices, demonstrating a significant improvement in subject clustering for the tangent FCs in comparison to the ones lying on the manifold. Notably, subject 8, whose data seems corrupted in the original space, is correctly clustered when projected into tangent space.

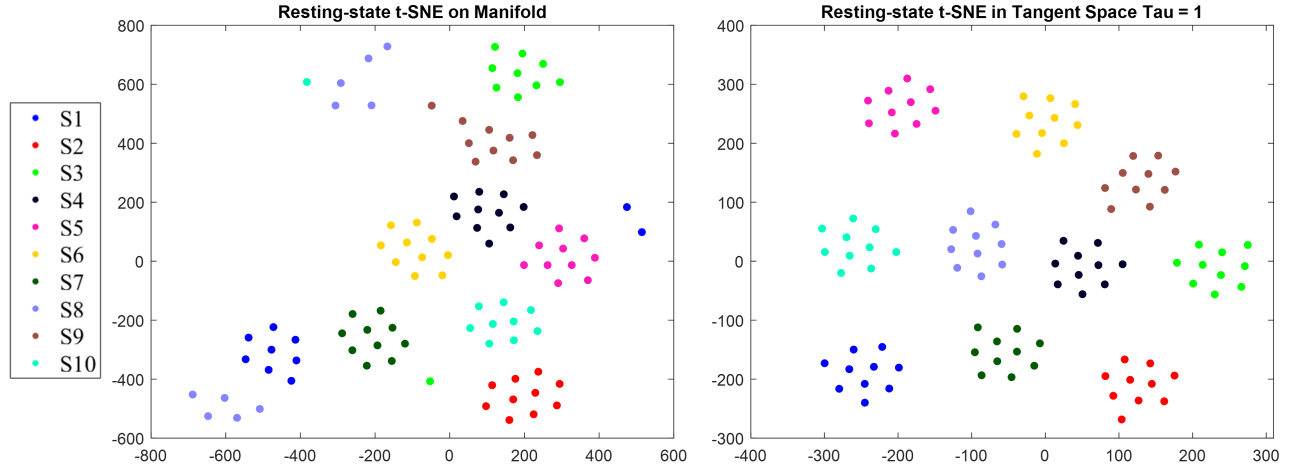


Figure 3.3. t-SNE visualization of manifold and tangent meta identifiability matrices for resting-state and regularization value of 1. Visually, clustering results are shown to be significant better in tangent space in comparison to results from manifold FCs.

3.4 Discussion

Figure 3.3 presents a comparison between original and tangent-space meta-identifiability matrices using Pearson correlation and Euclidean distance. The enhanced separability observed in tangent-space matrices underscores the benefits of this transformation for FC fingerprinting. Pearson correlation particularly improves distinctiveness by amplifying subject-specific connectivity patterns. These findings support the hypothesis that tangent space projection enhances the discriminability of FC fingerprints across scanning sessions, corroborating previous research on FC manifold representations.

Figure 3.3 showcases the clustering of subject-specific FCs using t-SNE, comparing the original and tangent-space meta-identifiability matrices. The clearer clustering in tangent space confirms that this representation better preserves individual connectivity patterns

across sessions. The ability of t-SNE to differentiate subjects more effectively in tangent space aligns with findings from studies that employ non-linear dimensionality reduction techniques for neuroimaging data. Notably, the recovery of previously corrupted sessions further highlights the robustness of this approach.

Despite these promising results, several limitations exist. First, the reliance on a reference FC matrix for tangent space projection may introduce biases if not appropriately selected. Future research should explore adaptive strategies for defining the reference matrix to optimize subject-specific representation. Additionally, the computational demands of tangent space projection present challenges for large-scale neuroimaging datasets, necessitating further optimization of these methods.

3.5 Conclusion

This study demonstrates the effectiveness of Riemannian geometry-based approaches in enhancing functional connectome fingerprinting. By leveraging tangent space projection, we improved the distinctiveness of individual functional connectivity (FC) patterns, addressing limitations of traditional Euclidean-based comparisons. The findings indicate that tangent FCs provide higher identifiability rates across multiple scanning sessions, reinforcing the hypothesis that mapping FCs onto a Euclidean space via tangent space projection facilitates more stable and robust subject-specific characterizations.

Additionally, the application of t-Stochastic Neighbor Embedding (t-SNE) to tangent FCs revealed improved clustering of individual connectivity profiles, highlighting the geometric advantages of the tangent space representation. These results align with prior research demonstrating the benefits of Riemannian geometry in neuroimaging applications and suggest that incorporating manifold-based approaches can significantly enhance the accuracy of FC fingerprinting.

Despite these promising findings, certain limitations remain. The computational cost associated with tangent space projections, as well as potential sensitivity to hyperparameter selection in distance metrics, require further investigation. Future work should explore hybrid

models that integrate tensor decomposition techniques with Riemannian geometry-based frameworks to optimize fingerprinting performance while maintaining interpretability.

4. CONCLUSION

The presented chapters contribute to advancing the field of FC fingerprinting by demonstrating the efficacy of different dimensionality reduction techniques. The Tucker decomposition-based framework presented in the first chapter showcases the potential of tensor methods in extracting individual-specific connectivity patterns, improving fingerprinting accuracy across various fMRI conditions for both within- and between-condition settings. Meanwhile, the work presented in the second chapter demonstrates that tangent space projection offers a geometric perspective that enhances the robustness of FC comparisons. Collectively, these approaches highlight the importance of integrating advanced mathematical frameworks into neuroimaging studies to improve the reliability and reproducibility of individual brain fingerprinting. Future research can build upon these findings by exploring hybrid techniques that combine tensor decomposition with geometric representations, further refining the characterization of functional brain networks.

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